

Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

Lab 3: Data Cleaning - Handling Missing Values

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Problem Statement 1

A residential housing dataset is provided in a CSV file (`residential_housing.csv`). The dataset contains various attributes that describe the physical, locational, and quality-related characteristics of houses. These features are:

Table 1: Description of Attributes in the Cleaned Residential Housing Dataset

Feature	Description
Id	Unique identifier assigned to each property record.
MSSubClass	Type of dwelling involved in the transaction.
MSZoning	General zoning classification of the property.
LotFrontage	Linear feet of street connected to the property.
LotArea	Total lot size in square feet.
Street	Type of road access to the property.
Alley	Type of alley access to the property.
Utilities	Type of utilities available at the property.
LandSlope	Slope of the property.
Condition2	Secondary proximity to main road or railroad.
BldgType	Type of dwelling.
OverallQual	Overall material and finish quality.
OverallCond	Overall condition rating.
YearBuilt	Original construction year.
YearRemodAdd	Year of remodeling or addition.
RoofMatl	Roof material used.
MasVnrArea	Masonry veneer area in square feet.
ExterQual	Exterior material quality rating.
Foundation	Type of foundation.
BsmtQual	Basement height quality.
BsmtCond	Basement general condition.
BsmtExposure	Basement exposure level.
BsmtFinSF1	Size of primary finished basement area.
BsmtFinSF2	Size of secondary finished basement area.

Feature	Description
BsmtUnfSF	Unfinished basement area.
TotalBsmtSF	Total basement area.
Heating	Type of heating system.
CentralAir	Availability of central air conditioning.
Electrical	Electrical system type.
1stFlrSF	Area of the first floor in square feet.
2ndFlrSF	Area of the second floor in square feet.
LowQualFinSF	Low-quality finished living area.
GrLivArea	Above-ground living area.
BsmtFullBath	Number of basement full bathrooms.
BsmtHalfBath	Number of basement half bathrooms.
FullBath	Number of full bathrooms above ground.
HalfBath	Number of half bathrooms above ground.
BedroomAbvGr	Number of bedrooms above ground.
KitchenAbvGr	Number of kitchens above ground.
TotRmsAbvGrd	Total number of rooms above ground.
Fireplaces	Number of fireplaces.
FireplaceQu	Fireplace quality rating.
GarageType	Garage location type.
GarageYrBlt	Year garage was built.
GarageCars	Number of cars the garage can hold.
GarageArea	Size of garage in square feet.
GarageQual	Garage quality rating.
WoodDeckSF	Wood deck area in square feet.
OpenPorchSF	Open porch area in square feet.
EnclosedPorch	Enclosed porch area.
3SsnPorch	Three-season porch area.
ScreenPorch	Screen porch area.
PoolArea	Pool area in square feet.
PoolQC	Pool quality rating.
MiscVal	Value of miscellaneous feature.
MoSold	Month of sale.
YrSold	Year of sale.
SaleType	Type of sale.
SaleCondition	Condition of sale.
SalePrice	Recorded sale price of the property (target variable).

Write a Python program to perform the following tasks:

1. Display the number of missing values in each attribute. Also, find the total number

of missing values in the file.

2. For the file `residential_housing.csv`:

- (a) Count and display the number of tuples having one, two, three, four, or upto nine missing values. Plot a graph for the `number of missing values` vs. the `number of tuples`.
- (b) Count and display the number of tuples having equal to or more than 50% of attributes with missing values.
- (c) Delete (drop) the tuples equal to or more than 50% of attributes with missing values.
- (d) The target (class) attribute is `SalePrice`. Drop the tuple missing in the target (class) attribute.
- (e) Then save cleaned dataset after doing steps (c) and (d) as `residential_housing_updated.csv`

Instruction: Create a new Jupyter Notebook and load the updated CSV file obtained after deleting the required tuples in the previous step. All further analysis must be performed using this updated dataset.

3. Count and display the number of missing values in each attribute. Also, find the total number of missing values in the file (after the deletion of tuples).

4. For the file `residential_housing_updated.csv` - Experiments on Filling Missing Values

- (a) Compute the mean, median, mode, and standard deviation for each numerical attribute for the file `residential_housing_updated.csv`.
- (b) Replace the missing values for numerical attributes with the median of their respective attributes. (Use `df.fillna()` with suitable arguments.) Count and display the number of missing values in each attribute after replacing missing values.
- (c) Compute the mean, median, mode, and standard deviation for each numerical attribute after replacing the missing values with median and observe the changes with respect to step (a).
- (d) Replace the missing values in each numerical attribute by propagating the previous non-missing value (forward padding). (Use `df.fillna()` with suitable arguments). Count and display the number of missing values in each attribute after replacing missing values.
- (e) Replace the missing values in each numerical attribute by propagating the next non-missing value (backward padding). (Use `df.fillna()` with suitable arguments). Count and display the number of missing values in each attribute after replacing missing values.
- (f) Replace the missing values in each numerical attribute using the linear interpolation technique. (Use `df.interpolate()` with suitable arguments). Count and display the number of missing values in each attribute.